

RARAU, Romania

WMO: 150520

Lat: 47.4496N

Lon: 25.5680E

Elev: 5164

StdP: 12.15

Time Zone: 2.00 (EUE)

Period: 90-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.8	2.9	-10.2	3.8	10.6	-2.6	5.8	8.9	43.8	22.2	32.1	21.0	7.9	0	0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	69.3	58.2	66.0	57.5	63.2	56.1	60.5	65.6	58.9	63.8	57.3	61.6	4.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.8	89.9	62.4	57.1	84.5	60.5	55.6	80.0	59.2	29.4	65.9	28.1	63.8	27.0	61.6	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.6	23.5	21.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-6.4	65.4	6.9	3.6	-11.4	68.0	-15.4	70.1	-19.2	72.1	-24.2	74.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	21.2	21.3	24.7	34.0	44.3	50.7	53.7	53.6	44.8	38.8	28.5	21.4		
	DBStd	14.83	9.15	9.48	9.55	7.86	7.49	6.35	5.80	6.62	7.98	9.34	9.86	7.96		
	HDD50	5390	891	803	786	481	203	69	31	37	198	358	646	887		
	HDD65	10407	1356	1223	1251	931	641	431	353	355	606	812	1096	1352		
	CDD50	462	0	0	0	0	27	89	144	149	42	11	0	0		
	CDD65	4	0	0	0	0	0	0	1	2	1	0	0	0		
	CDH74	10	0	0	0	0	0	2	3	3	2	0	0	0		
	CDH80	1	0	0	0	0	0	0	0	0	1	0	0	0		
(14) Wind	WSAvg	6.3	6.8	7.8	8.1	6.6	6.1	4.9	4.6	4.1	6.0	6.1	6.9	7.5		
(15) Precipitation	PrecAvg	28.8	1.0	1.2	1.5	2.2	3.6	4.4	4.5	3.3	2.6	2.0	1.4	1.4		
	PrecMax	43.3	1.8	2.4	2.9	4.3	6.6	9.1	9.1	5.0	6.9	5.0	2.8	2.6		
	PrecMin	19.9	0.2	0.3	0.4	0.9	1.7	2.4	2.0	1.8	0.9	0.2	0.0	0.4		
	PrecStd	5.0	0.4	0.4	0.7	0.9	1.5	1.7	2.0	0.9	1.3	1.1	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	44.1	51.2	54.0	65.6	70.2	71.7	74.1	69.7	61.7	51.9	42.8		
		MCWB	31.4	35.1	41.6	46.4	52.6	59.4	59.1	61.0	56.2	53.0	43.3	36.4		
	2%	DB	38.7	39.5	45.2	49.9	60.4	65.5	68.2	69.7	62.9	56.6	47.7	35.7		
		MCWB	30.9	33.6	38.0	44.0	50.5	57.9	59.1	58.4	55.2	49.2	41.5	32.9		
	5%	DB	35.5	36.4	41.1	47.3	57.4	62.4	64.8	66.1	59.6	53.8	44.7	33.7		
		MCWB	30.6	32.2	36.4	42.2	50.1	55.0	57.1	58.0	53.2	47.8	41.1	32.8		
	10%	DB	32.9	33.6	37.7	44.3	54.4	59.5	62.4	62.7	55.6	50.9	42.0	31.7		
		MCWB	30.4	30.8	34.6	40.2	48.6	54.2	56.5	56.5	51.4	46.8	38.7	30.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	36.8	42.5	48.5	55.2	61.3	63.9	63.5	59.3	54.0	47.2	36.8		
		MCDB	37.8	39.8	47.3	52.1	59.2	67.8	67.5	68.8	64.1	58.0	48.2	39.3		
	2%	WB	33.6	34.9	39.1	45.0	52.6	59.0	60.3	60.7	56.3	51.3	43.5	34.7		
		MCDB	35.1	38.0	42.9	48.2	57.9	63.9	64.2	66.4	61.2	54.3	45.6	36.0		
	5%	WB	31.9	32.9	36.8	42.6	50.9	56.9	58.7	58.8	54.0	49.6	41.2	33.0		
		MCDB	33.6	35.7	39.8	46.5	55.8	60.8	63.0	64.1	58.5	52.3	43.3	33.5		
	10%	WB	30.4	31.1	34.6	40.7	49.3	55.1	57.1	57.1	51.9	47.3	39.2	30.8		
		MCDB	32.2	33.1	38.0	43.7	53.5	58.5	61.3	61.3	55.3	50.0	41.7	31.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.5	10.7	10.3	11.6	11.5	11.5	11.3	10.8	11.4	9.4	9.0		
		MCDBR	10.1	13.2	15.4	13.4	15.8	15.3	16.0	15.5	15.4	14.5	10.8	10.0		
	5% WB	MCWBR	7.6	9.8	9.8	8.5	9.0	9.2	9.0	8.5	9.8	9.5	8.4	8.2		
		MCWBR	9.1	11.9	14.0	12.2	12.9	14.4	14.0	13.6	13.7	12.1	9.9	9.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.253	0.277	0.310	0.357	0.342	0.340	0.359	0.363	0.329	0.301	0.278	0.255		
		taud	2.426	2.386	2.382	2.316	2.408	2.428	2.371	2.360	2.438	2.513	2.526	2.475		
	Ebn at Noon	Ebn at Noon	271	285	290	284	292	292	285	279	280	273	259	257		
		Edh at Noon	21	27	32	38	36	35	37	36	30	24	19	18		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	380	640	987	1324	1627	1728	1721	1590	1136	767	430	294		
	RadStd	RadStd	57	76	128	132	165	144	161	146	147	106	60	41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	+1.28	N/A	N/A	+1.71	N/A	N/A	-283	-396	+185	+63

Nomenclature: See separate page